

Lovepreet Singh

LinkedIn | GitHub | Portfolio | LeetCode | Codeforces

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EDUCATION

- Dr. B.R. Ambedkar National Institute of Technology (NIT), Jalandhar** Jalandhar, Punjab
B.Tech in Computer Science & Engineering; CGPA: 8.27/10 Aug 2024 – May 2028

EXPERIENCE

- Google Developer Student Club (GDSC), NIT Jalandhar** Dec 2024 - Present
Software Development & Competitive Programming Core Team Co-Lead

PROJECTS

- Distributed Network File System (NFS)** May 2026 - Jun 2026
C, TCP/IP Sockets, POSIX Threads, LRU Cache, Linux Systems | GitHub
 - Distributed Architecture:** Architected a distributed Network File System in **C**, leveraging **TCP/IP sockets** and **POSIX threads** to coordinate communication between clients, a Naming Server, and **5+ Storage Servers**.
 - Data Structures & Concurrency:** Accelerated path resolution by **40%** using a custom **Trie** index and **LRU Cache**; implemented readers-writer locks for safe concurrent multi-client file operations.
 - Fault Tolerance & I/O:** Devised resilient storage protocols featuring asynchronous buffered writes, automated replication across **3 storage nodes**, and heartbeat-based failure detection for high availability and data consistency.
- ArthMitra | AI-Powered Personal Finance Assistant** April 2026 - May 2026
LangGraph, LangChain, Python, Supabase (PostgreSQL), ChromaDB, Streamlit | GitHub
 - Multi-Agent System:** Orchestrated a **LangGraph**-powered agentic architecture, utilizing a Router Agent to dynamically classify user intents across **5 specialized LLM agents** alongside a persistent, state-monitoring Alert Agent.
 - RAG Pipeline & Vector Search:** Established a **Retrieval-Augmented Generation (RAG)** framework via **LangChain** and **ChromaDB**; leveraged **vector embeddings** to ground AI financial advice across **1000+ transaction records** securely in the user's historical data.
 - Automated ETL & Database:** Constructed an automated **Python** pipeline for parsing and categorizing PDF bank statements; designed a scalable **PostgreSQL (Supabase)** schema supporting **10k+ transaction entries**.
- Campora | MERN + Socket.IO Event & Club Management Platform** Jan 2026 - April 2026
MongoDB, Express.js, React, Node.js, Socket.IO, Tailwind CSS, Zustand | GitHub | Live
 - Scalable Architecture:** Developed a production-grade MERN platform with modular MVC architecture for managing university clubs, events, and announcements serving **500+ users**.
 - Real-Time & Access Control:** Integrated **WebSocket-based** real-time notifications via Socket.IO and established a **4-tier Role-Based Access Control (RBAC)** system using JWT authentication and middleware-enforced authorization.
 - Deployment:** Provisioned the application on **Render** with **MongoDB Atlas**; accelerated REST API response times by **30%** through efficient query indexing and database optimization.

SKILLS

- Languages:** C++, Python, JavaScript, SQL, C, Bash
- Web & Frameworks:** React.js, Node.js, Express.js, Socket.IO, REST APIs, JWT, Streamlit
- Databases & Cloud:** MongoDB, PostgreSQL, MySQL, Firebase, Supabase, MongoDB Atlas, Render, Vercel
- ML/AI Tools:** LangGraph, LangChain, LLM APIs (OpenAI/Anthropic), ChromaDB, Vector Embeddings, RAG, Prompt Engineering
- CS Fundamentals:** CS Fundamentals: Data Structures Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, System Design
- Tools:** Git, GitHub, Linux, Figma, Postman

ACHIEVEMENTS

- Data Structures & Algorithms:** Attained Codeforces **Pupil** (1200+ rating); Solved **800+ algorithmic challenges** (LeetCode, GeeksforGeeks, CodeStudio), specializing in Dynamic Programming, Graph Theory, and space-time complexity optimization.
- Secured 3rd place among 100+ participants** in CodeStorms 2025, a campus-wide algorithmic coding contest.
- AIR 14,992 (99.07 Percentile):** Ranked among the top **1% of 1.23 million+ candidates** nationwide in JEE Main 2024.